11 September 2025

Comprehensive synthesis of K48/K63 ubiquitin pentamers by graph-empowered automation

Sohei Majima[1], Eric Kummelstedt[1], Toshiki Mikami[1], Leonardo Seidl[1], Jiling Han[1], Jeffrey Bode[1]

1. ETH Zurich

Abstract

Linear, branched, and heterotypic ubiquitin (Ub) chains regulate essential cellular processes, yet their structural and functional analysis remains limited by the difficulty of preparing defined, homogeneous samples by chemical synthesis or biochemical assembly. We established a fully automated platform for the comprehensive synthesis of all 56 K48/K63-linked Ub tetramers and pentamers. This is achieved by iterative cycles of K48 or K63-specific chemoenzymatic conjugations of chemically synthesized Ub monomers bearing a new, fast-cleaving lysine protecting group compatible with folded proteins. A graph-based digital encoding of the Ub chain topologies enabled systematic enumeration and filtering of reaction pathways, guiding the selection of viable synthetic routes. This computational system generates reagent maps and executable code to direct iterative cycles of deprotection and conjugation on a standard, laboratory liquid handling robot. All 14 Ub tetramers and 42 Ub pentamers were obtained with high fidelity, as validated by SDS-PAGE and mass spectrometry. This work establishes a general framework for automated protein conjugate synthesis and provides on-demand access to the complete space of K48/K63 branched ubiquitin chains for biological and structural studies.

Keywords

ubiquitin, bioconjugation, automation, chemoinformatics, digital chemistry

Posted on 11 September 2025 — CC-BY-NC 4.0 — This is a preprint and has not been peer reviewed. Data may be preliminary. —
https://doi.org/10.26434/chemrxiv-2025-943j7

Comprehensive synthesis of K48/K63 ubiquitin pentamers by graph-empowered automation

Sohei Majima[1], Eric Kummelstedt[1], Toshiki Mikami[1], Leonardo Seidl[1], Jiling Han[1], and Jeffrey W Bode[1]

[1]Laboratory for Organic Chemistry, Department of Chemistry and Applied Biosciences, ETH Zürich, Vladimir Prelog Weg 3, CH-8093 Zürich, Switzerland

[*] E-mail: bode@org.chem.ethz.ch

Abstract

Linear, branched, and heterotypic ubiquitin (Ub) chains regulate essential cellular processes, yet their structural and functional analysis remains limited by the difficulty of preparing defined, homogeneous samples by chemical synthesis or biochemical assembly. We established a fully automated platform for the comprehensive synthesis of all 56 K48/K63-linked Ub tetramers and pentamers. This is achieved by iterative cycles of K48 or K63-specific chemoenzymatic conjugations of chemically synthesized Ub monomers bearing a new, fast-cleaving lysine protecting group compatible with folded proteins. A graph-based digital encoding of the Ub chain topologies enabled systematic enumeration and filtering of reaction pathways, guiding the selection of viable synthetic routes. This computational system generates reagent maps and executable code to direct iterative cycles of deprotection and conjugation on a standard, laboratory liquid handling robot. All 14 Ub tetramers and 42 Ub pentamers were obtained with high fidelity, as validated by SDS-PAGE and mass spectrometry. This work establishes a general framework for automated protein conjugate synthesis and provides on-demand access to the complete space of K48/K63 branched ubiquitin chains for biological and structural studies.

https://doi.org/10.26434/chemrxiv-2025-943j7 **ORCID:** https://orcid.org/0000-0001-8394-8910 Content not peer-reviewed by ChemRxiv. **License:** CC BY-NC 4.0

Introduction

Ubiquitin (Ub) chains – generated by isopeptide linkages between the Ub C-terminus and one of seven lysine residues – are crucial for diverse aspects of cellular function, including protein homeostasis,[1–4] aggresome processing,[5–9] and autophagy.[10] The topological complexity of these biopolymers, which consist of covalently-linked Ub monomers, increases exponentially with chain length and branching. Pentamers constructed from just K48 and K63 branches comprise 42 distinct topologies, rising to 10,472 architectures if all seven lysines are considered.[11] Understanding the biological functions, binding partners, and structural consequences of these chain topologies requires access to homogeneous materials – a challenge that remains largely unmet using existing biochemical or synthetic approaches.[12] To date, only three homogeneous Ub pentamers of any type have been prepared.[13,14]

As with other biopolymers – including proteins, nucleic acids, and carbohydrates – the multitude of possible sequences necessitates modular and automated synthetic approaches from defined building blocks. In general, implementing automated synthetic platforms requires three key components: iterative coupling and deprotection reactions, robotic systems for autonomous operation, and a digital platform to encode the biopolymers and control the robotic devices. While such systems are well-established for the preparation of DNA oligomers and peptides[15,16] – and emerging for carbohydrates and small molecules[17–23] – no equivalent platform exists for assembling oligomeric protein–protein conjugates such as Ub chains. Here we present a modular, automated, and computationally-guided platform for the comprehensive synthesis of all 56 possible K48/K63-linked ubiquitin tetramers and pentamers (Figure 1). This platform integrates several core technologies: 1) systematic encoding of all possible Ub chains as a machine-readable graph, 2) transient and permanent lysine protecting groups compatible with folded Ub chains, 3) scalable, modular syntheses of the requisite, selectively masked Ub monomers, 4) computational enumeration and selection of optimal synthetic routes to all possible K48/K63 tetramers and pentamers, and 5) automated generation of source plate maps and scripts for executing automated chain synthesis on a low-cost, open-source liquid handling robot. The current setup can synthesize up to 14 discrete Ub pentamers in parallel using small liquid-handling robots already present in many academic and industrial labs.

The core challenge in Ub chain synthesis is site-selective isopeptide bond formation between folded protein monomers at specific lysine residues. In natural systems, this is orchestrated by sophisticated protein assemblies that control linkage specificity.[24,25] Current laboratory approaches to branched or heterotypic ubiquitin chains rely on enzymatic, chemical, or hybrid approaches.

https://doi.org/10.26434/chemrxiv-2025-943j7 **ORCID:** https://orcid.org/0000-0001-8394-8910 Content not peer-reviewed by ChemRxiv. **License:** CC BY-NC 4.0

Enzymatic methods employing E2 or E3 enzymes in combination with Ub monomers bearing Lys–>Arg mutations have been used to construct defined architectures, such as Schulman's and Kulathu's K48/K63 chain building.[13,26] Total chemical synthesis provides precision access to homogenous Ub oligomers, most notably K48 tetramers from Brik[27] and K11/K48 branched chains from Liu and co-workers.[28] Hybrid approaches combining chemically synthesized, selectively protected Ub monomers with enzymatic conjugation – as reported by Okamoto[29] and our lab[14] – offer greater flexibility. Their current implementations, however, are labour-intensive and require long deprotection times that preclude automated synthesis of diverse chain topologies.

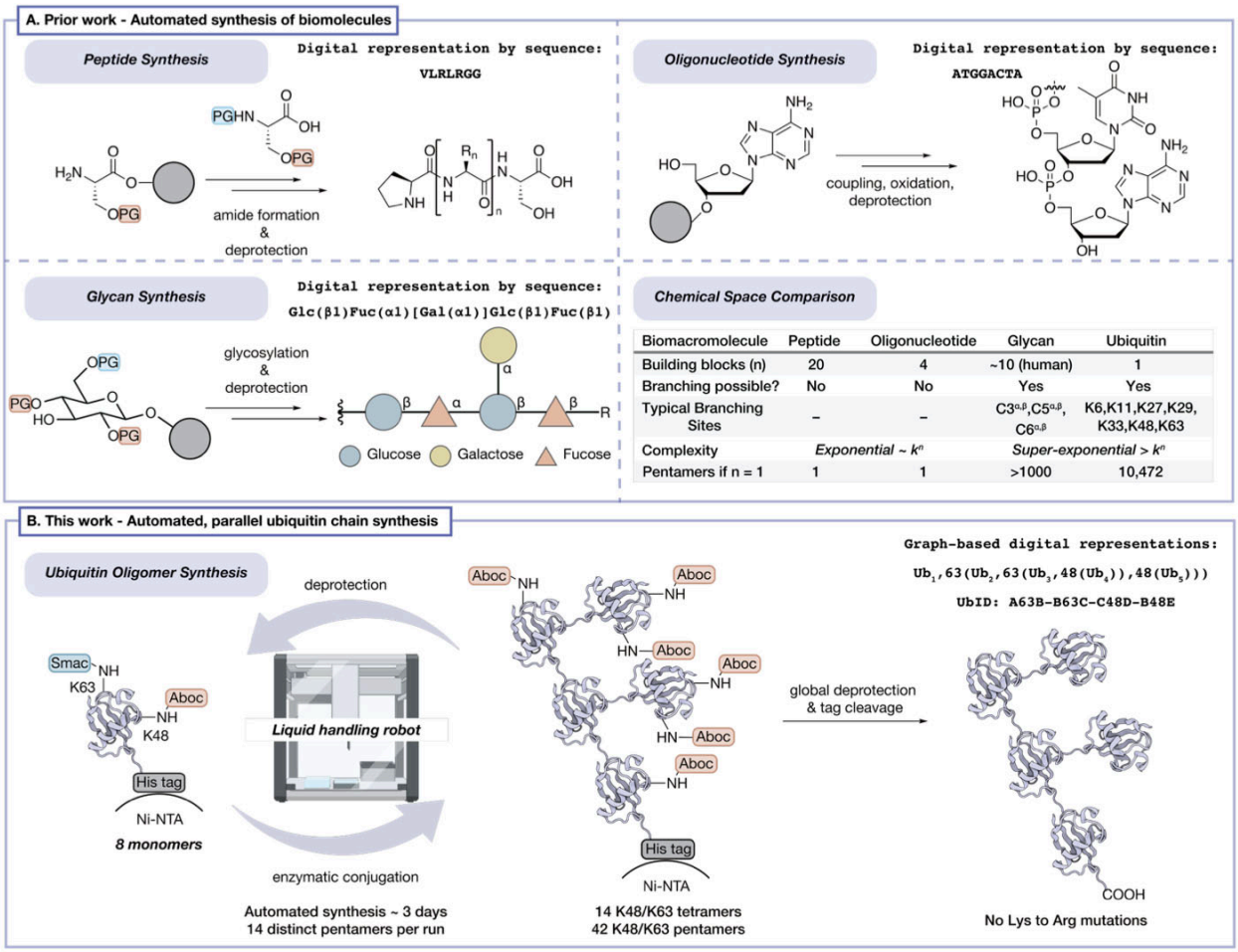


Figure 1. Automated synthesis platforms for biomolecules. A) Established platforms for peptide, nucleic acid, and carbohydrate synthesis rely on iterative cycles of monomer coupling and deprotection, guided by digitally encoded representations of the molecules to enable computer control of a robotic platform. This is straightforward for linear chains, but becomes increasingly complex – even for a single building block – with branched architectures. B) This work – graph-based digital representations of Ub chains enable automated, iterative chemoenzymatic conjugations and lysine deprotections on robotic liquid handling platform for the comprehensive synthesis of all 56 K48/K63-linked Ub tetramers and pentamers.

https://doi.org/10.26434/chemrxiv-2025-943j7 **ORCID:** https://orcid.org/0000-0001-8394-8910 Content not peer-reviewed by ChemRxiv. **License:** CC BY-NC 4.0

Results

Our automated chemoenzymatic synthesis relies on iterative cycles of chemical deprotections and chemoenzymatic Ub donor conjugations. A bottleneck in our previously established proof-of-principle approach was the use of the Abac (aminobutanamide-derived carbamate) lysine protecting group, which required 16 h at 37 °C for pyridoxal phosphate (PLP)-mediated deprotection, limiting full automation.[14] To address this, we developed a new PLP-cleavable group, Smac (**s**erine **m**ethyl**a**mide), which is removed within 2 h at slightly basic pH at 37 °C (Figure 2A). It remains compatible with our previously reported orthogonal Aboc protecting group (cleaved with $NaIO_4$), used to protect lysines throughout chain elongation.

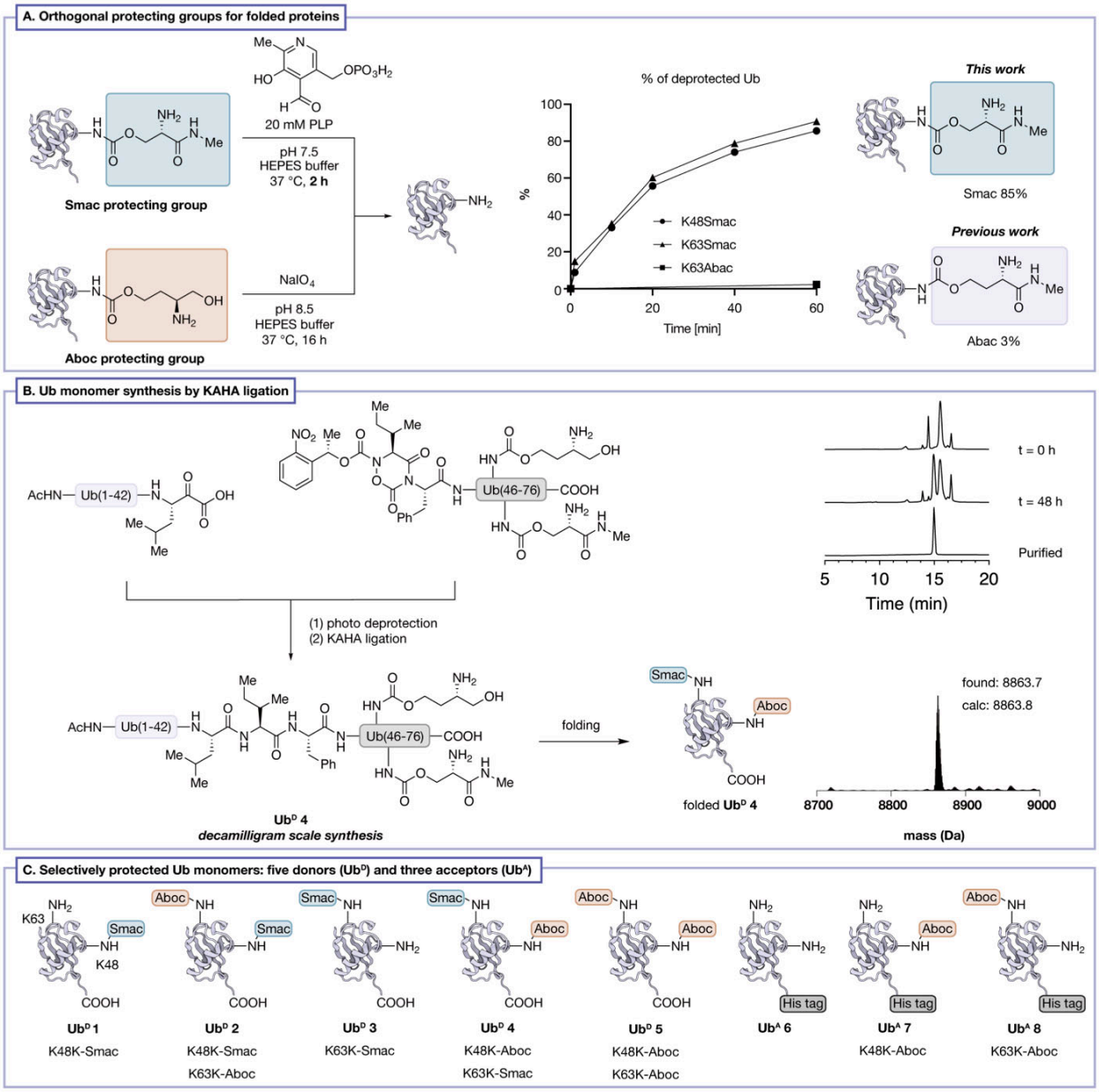


Figure 2. Overview of the Ub chain building strategy. A) The structure of and deprotection kinetics of Smac and Aboc protecting group (20 mM PLP, 200 mM HEPES pH 8, 150 mM NaCl). B) Representative synthesis of Ub monomer **Ub^D 4** with Smac and Aboc group by KAHA ligation. C) The eight selectively protected Ub donors and acceptors employed for automated synthesis of Ub chains. His tag represents an additional DH_6 sequence at the C-terminus.

https://doi.org/10.26434/chemrxiv-2025-943j7 **ORCID:** https://orcid.org/0000-0001-8394-8910 Content not peer-reviewed by ChemRxiv. **License:** CC BY-NC 4.0

With the protecting groups in hand, we proceeded to the synthesis of the Ub monomers (Figure 2B). To improve synthetic efficiency and overall yield of Ub monomers, we split the sequence into two unprotected peptide segments joined between Leu43 and Ile44 by a KAHA ligation[30] employing a cyclic dipeptide hydroxylamine that affords a scarless ligation site.[31] This approach proved more scalable and reproducible than single chain synthesis[32] and reliably delivered decamilligram quantities in high purity after a single HPLC purification. Protecting groups at K48 and K63 were introduced in segment 2 (residues 44–76), which was coupled to a constant segment 1 (residues 1–43). We prepared two classes of Ub monomers: Ub acceptors (**Ub^{A}**), which are C-terminally His tagged to allow Ni-NTA immobilization and prevent undesired conjugation[33]; and Ub donors (**Ub^{D}**), which contain selectively protected lysines to block uncontrolled chain extension. In total, we prepared eight distinct Ub monomers – five **Ub^{D}**s and three **Ub^{A}**s (Figure 2C) – which together constitute the complete set of building blocks for automated synthesis of K48/K63-linked Ub chains.

For lysine selective chemoenzymatic conjugation of Ub monomers, we employed cocktails of Uba1 (Ubiquitin-activating enzyme E1), E2 ubiquitin-conjugating enzymes, and the appropriate **Ub^{D}**. For K63 chain elongation, the established Ubc13/Mms2 E2 was used.[34] K48 attachment was achieved with a fusion of the gp78 RING domain with Ube2g2, as reported by Blythe *et al*.[35,36] Ube2K, although canonically K48-selective, was restricted to K48 conjugations onto a K63-linked Ub due to observed promiscuity during elongation, consistent with its known affinity for K63-linked substrates.[37] For the automated synthesis platform, the appropriate E2 was paired with one of the five **Ub^{D}** monomers, along with Uba1 and ATP, resulting in 12 distinct conjugation mixtures.

Establishing a fully automated synthesis platform for Ub chains requires a digital representation of Ub sequence and topology for synthetic planning, encoding of the intermediates, and ensuring that our monomer-enzyme combinations can deliver all possible Ub chain topologies. Recognizing that ubiquitin chains possess an inherently graph-like structure, we developed a formal graph framework ubiG(V,E) – **ubi**quitin **g**raph – in which vertices (V) represent individual Ub monomers and edges (E) the lysine linkage sites.[38] The vertices (V) are numbered by pre-order traversal, enabling a standardized and biologically consistent indexing scheme (see Figure 3A and Extended Data Figure 1).[39] These graphs were encoded using JavaScript Object Notation (JSON), chosen for its inherent support for rooted and nested structures and compatibility with automation tools (see Extended Data Figure 2).[40]

Building on this foundation we systematically enumerated all theoretically accessible lysine-linked ubiquitin multimer configurations up to the pentamer level for K6, K11, K27, K29, K33, K48, and K63 branches. This effort generated a comprehensive JSON database, yielding 819 distinct tetrameric[41] and 10,472 pentameric configurations explorable through a custom application

https://doi.org/10.26434/chemrxiv-2025-943j7 **ORCID:** https://orcid.org/0000-0001-8394-8910 Content not peer-reviewed by ChemRxiv. **License:** CC BY-NC 4.0

supporting five different graph-based nomenclature systems.[38,42] When restricted to the K48/K63 linkages, the resulting subset yielded 14 tetramers and 42 pentamers.

Within the ubiG framework, each deprotection and conjugation reaction is encoded as transformation function that operates on the graph to generate product ubiGs (Figure 3B and Extended Data Figure 2). These functions, iteratively executed within a loop, form a simulation engine analogous to a breadth-first search (BFS), enabling exhaustive exploration of all feasible sequences compatible with the defined K48/K63-linked topologies (Figure 3C).[43]

To restrict the number of synthetic paths and ensure compatibility with our experimental setup, we implemented several design constraints. These included: 1) eliminating reactions that could lead to polymerization, such as **Ub^{D} 3** (K63K-Smac) with Ube2K; 2) removing reactions that left the material unchanged (i.e. deprotections or conjugation step that produced no structural change), and 3) excluding reactions that result in simultaneous conjugation of multiple ubiquitin monomers, such as concurrent reactivity at multiple unprotected K63 sites (Figure 3D). Notably, detecting structural transformations in the pre-order numbering within reactant and product ubiGs enabled automated enzyme selection: Ubc13/Mms2 for K63 addition, gp78-Ube2g2 for K48 chain extension and Ube2K for K48 branching (see Extended Data Figure 3). For each multimeric level, 448 and 2,168 viable routes were identified. A more refined subset was generated by filtering out end products retaining the temporary Smac protecting groups in favor of Aboc protected K48/K63. This yielded a manageable 26 and 114 synthetic routes for 14 tetrameric and 42 pentameric configurations (Figure 3D).

Intriguingly, several non-obvious facets of this approach to Ub chain synthesis emerged. First, all 42 K48/K63 pentamers are accessible within the 114 optimal pathways, confirming that our monomer selection, conjugation conditions, and protecting group strategy provides comprehensive access to all possible Ub tetramers and pentamers. Second, a histogram analysis of the Ub topologies encoded by the optimized routes allows broad classification of prevalence as a function of chain type. All linear and heterotypic chains have only a single viable synthetic route; all branched topologies can be access by at least two distinct pathways (Figure 3D). In choosing routes for automation, branching pathways initiated by K63 elongation or at later stages were preferred.

https://doi.org/10.26434/chemrxiv-2025-943j7 **ORCID:** https://orcid.org/0000-0001-8394-8910 Content not peer-reviewed by ChemRxiv. **License:** CC BY-NC 4.0

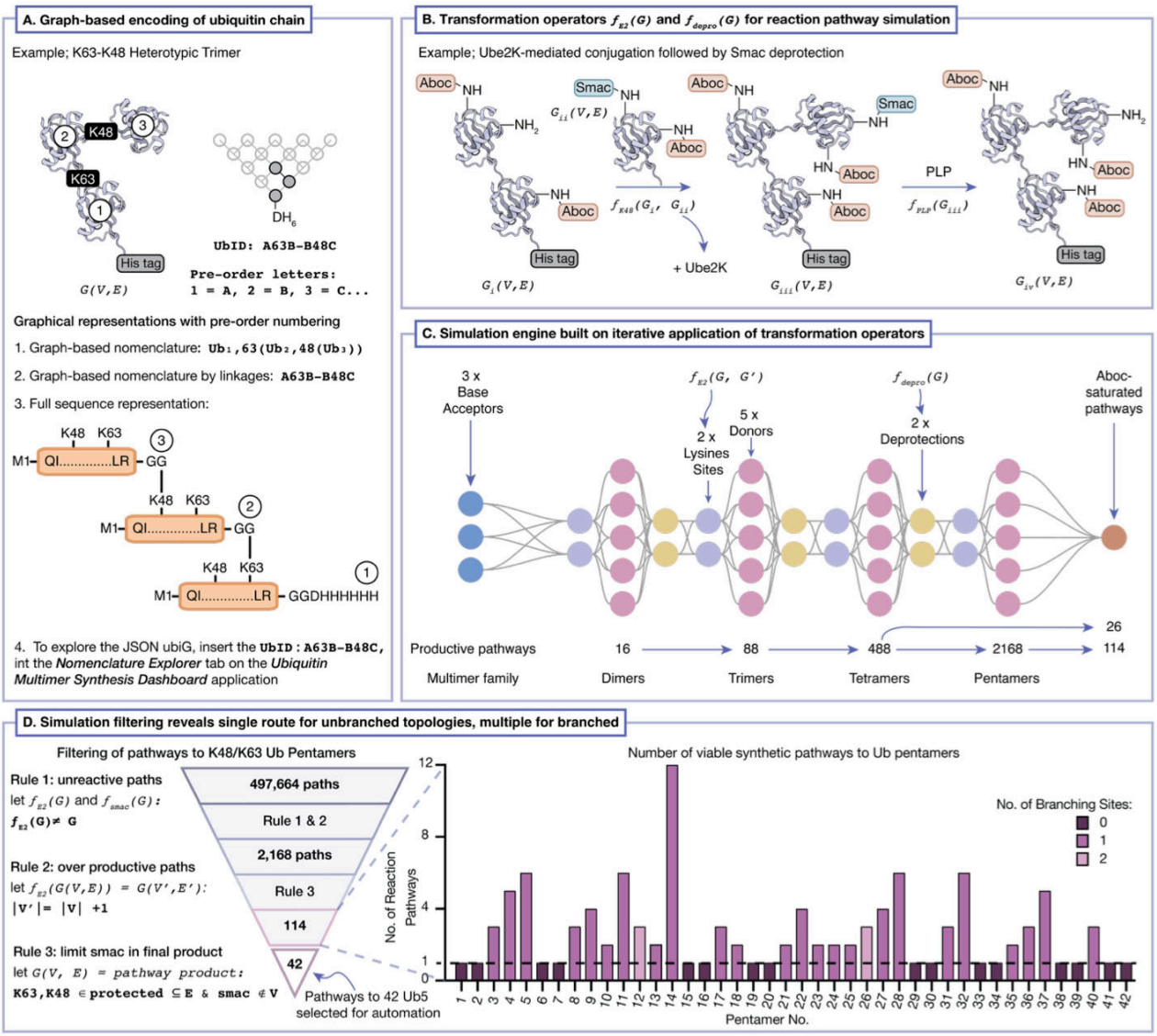


Figure 3. Computational simulation of Ub chains. A) All Ub chains — including those protected with Aboc, Smac, or unprotected— are represented within a machine-readable graphical framework *G(V,E)* termed ubiG. The graphs can be represented by several convenient nomenclature systems (see Extended Data Figure 1 and 'Nomenclature Explorer' tab of the downloadable application available on GitHub–https://github.com/erickummelstedt/ubiquitinformatics.git). B) This computable framework supports the application of defined functions, which fall into two reaction families: $f_{E2}(G, G_i)$ with twelve E2-mediated conjugations and $f_{depro}(G)$ with two deprotections. Each function encodes a chemically valid transformation which outputs a evolved *G(V,E)*. C) These functions form a simulation engine enabling the complete synthetic space to be mapped. D) The resulting set of valid pathways is compiled into a searchable reaction database, which is filtered to retain only those terminating in Aboc-saturated Ub chains species—yielding 26 routes for tetramers and 114 for pentamers. Among these 114 pathways, at least one exists for all 42 possible topologies.

With the synthetic landscape resolved, we set out to manifest the computationally selected routes to each Ub tetramer and pentamer into the physical reagent plates and dynamically generated control scripts for a small, widely-used liquid handling robot. To render the workflow compatible with a 96-well ANSI/SLAS format, several practical challenges had to be addressed: 1) efficient loading of Ub monomers onto resin, 2) establishing a modular workflow to support iterative incubations and washes, and 3) enabling parallelized conjugation and deprotection steps to reduce synthesis time.

https://doi.org/10.26434/chemrxiv-2025-943j7 ORCID: https://orcid.org/0000-0001-8394-8910 Content not peer-reviewed by ChemRxiv. License: CC BY-NC 4.0

The software platform integrates with the liquid handler by generating all necessary inputs for the selected Ub multimer targets. To accommodate limited deck space, we conducted the tetramer and pentamer syntheses from purified Ub dimers (Ub_2^A) immobilized on Ni-NTA resin at 10 nmol scale (Supplementary Information). The software platform generates two precise plate maps: one specifying the arrangement of the Ub_2^A on resins, and another detailing the twelve conjugation mixtures (E1 1 μM, E2 20 μM, **Ub^D** 100 μM [2 equiv relative to **Ub^A**], Mg-ATP 5 mM, TCEP 1 mM) (see Extended Data Figure 4). In addition, the interface generates a detailed reagent inventory, preparation instructions, and a customized Python (.py) script for robotic execution (Figure 4A).

The liquid handler deck is equipped with a cooler (CL), magnetic module (MM), and heater–shaker (HS), with additional positions allocated for reaction wash (RW), deprotection solution (DS), and deprotection wash (DW). Because PLP is unstable in solution, fresh PLP in degassed buffer was replenished prior to each deprotection step. Remaining deck space was reserved for pipette tips. The two plates, one containing the conjugation mixtures and the other **Ub^A** starting materials loaded onto magnetic Ni-NTA resin via their incorporated His tag, were respectively placed at the CL and MM positions (Figure 4A).

Figure 4B also illustrates the automated movement of solutions, plates and reagents throughout the four sequential operations – Deprotection, Wash, E2-Mediated Conjugation, Wash – that constitute one cycle of Ub addition. Each cycle required approximately 10 hours. Conjugation was performed by transferring conjugation mixtures onto **Ub^A**-loaded resin, incubating on the heater–shaker, and subsequently isolating the resin on the magnetic module before discarding the supernatant and washing. Smac deprotection followed an analogous sequence. Iterating these cycles enabled stepwise elongation of the chains, yielding Ub pentamers after three cycles.

https://doi.org/10.26434/chemrxiv-2025-943j7 **ORCID:** https://orcid.org/0000-0001-8394-8910 Content not peer-reviewed by ChemRxiv. **License:** CC BY-NC 4.0

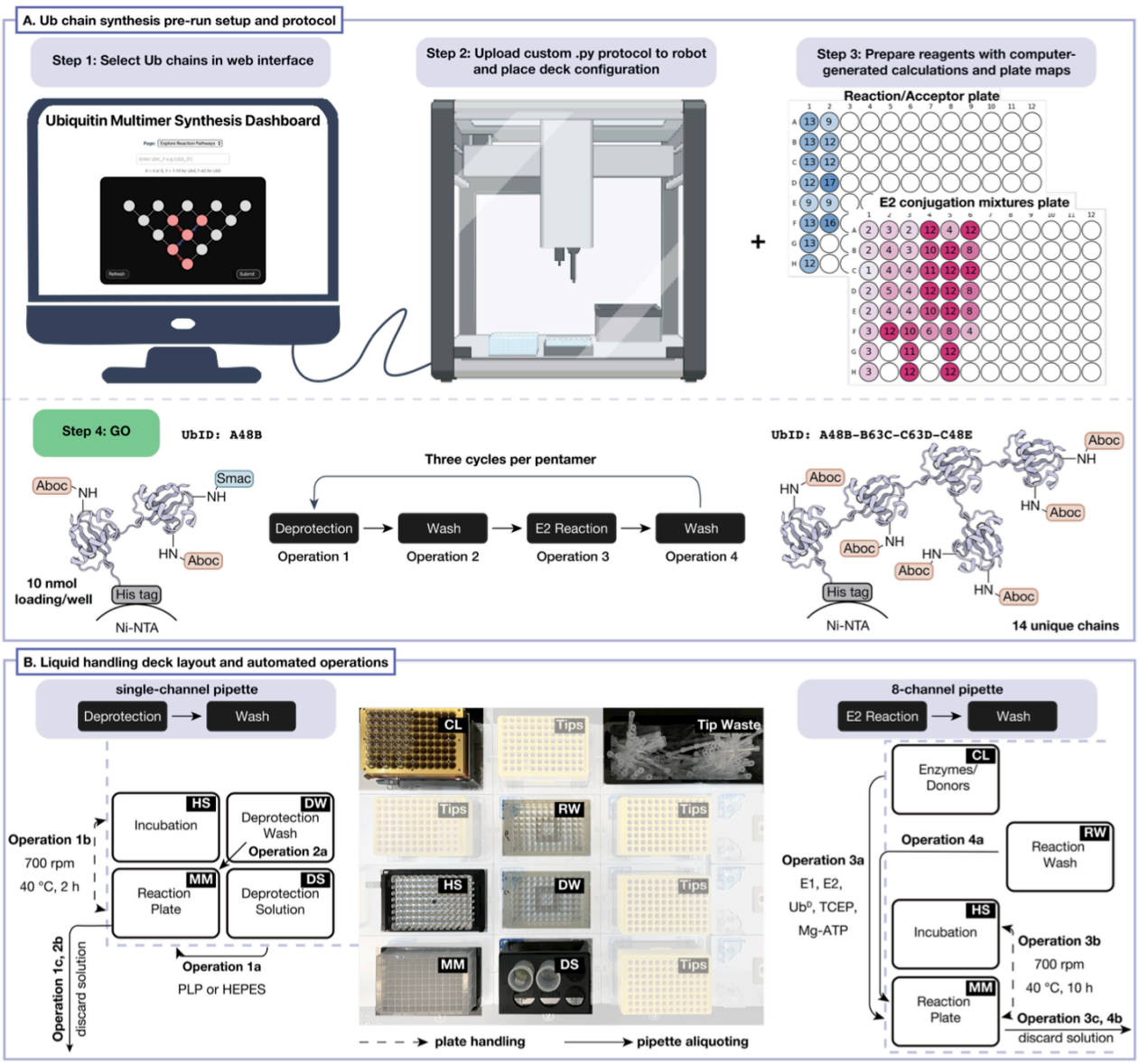


Figure 4. Overview of the automated synthesis of Ub chains. A) User interface of the Ub chain synthesis platform. The software digitally encodes selected chains and generates liquid handling operating code and annotated plate maps for reagent setup. B) Deck layout of the liquid handler showing modules for cooling (CL), heating/shaking (HS), magnetic separation (MM), and designated wells for deprotection, wash steps (RW, DW), and pipette tips. C) The transfer sequence for automated chain assembly. In each cycle, conjugation mixtures are transferred from the cooler (CL) to **Ub^{A}** on the MM. Conjugation reactions are incubated on the HS, followed by magnetic separation, waste removal, and washing. Smac deprotection uses the same sequence. Three consecutive cycles of these four operations produced Ub pentamers.

We validated this fully automated platform – which requires manual intervention only for preparation of the acceptor and source plates – by programming it to synthesize all possible K48/K63 linear, heterotypic, and branched Ub tetramers and pentamers. The 14 Ub tetramers included the two K48 and K63 linear chains, six heterotypic chains without branching, and six branched chains. Among these 12 heterotypic/branched Ub chains, five of them (**Ub$_4$ 3**, **4**, **9**, **12**, **13**) have been described previously[13,14,26,29], while the remaining seven are reported here for the first time. Following completion of the automated synthesis (2 cycles), the 14 tetramers were released from the Ni-NTA resin and the unpurified mixtures analyzed by SDS-PAGE (Figure 5A). Interestingly, each Ub chain

https://doi.org/10.26434/chemrxiv-2025-943j7 ORCID: https://orcid.org/0000-0001-8394-8910 Content not peer-reviewed by ChemRxiv. License: CC BY-NC 4.0

exhibits a distinct band position by Western blot analysis. This difference may arise from the structural characteristics of the linkages—K63-linked chains tend to adopt extended, linear conformations, whereas K48-linked chains are more compact and twisted.[1,11] During tetramer synthesis, the presence of higher molecular weight species, including pentamers, were occasionally observed. These side products were likely due to the promiscuous activity of Ube2K as well as inefficient wash steps, minor issues that were addressed prior to pentamer synthesis.

Automated synthesis of Ub pentamers was conducted in three batches of 14 pentamers each, in the arbitrary order shown in Figure 5B. All 42 K48/K63 pentamers, including two linear chains (**Ub_5 1** and **Ub_5 42**), the 14 heterotypic chains without branching, the 24 single-branched chains, and two doubly-branched chains (**Ub_5 12** and **Ub_5 26**) topologies were obtained on the first attempt. Western blot analysis of each reaction confirmed the presence of the bands corresponding to the expected pentamer, along with mass spectrometry consistent with the Aboc-protected, His-tagged products. Although topology-specific variances in the overall conversion and purity are apparent, the target Ub pentamers were the predominant species in all reaction sequences (see Supplementary Information for conversion data for all tetramers and pentamers).

https://doi.org/10.26434/chemrxiv-2025-943j7 **ORCID:** https://orcid.org/0000-0001-8394-8910 Content not peer-reviewed by ChemRxiv. **License:** CC BY-NC 4.0

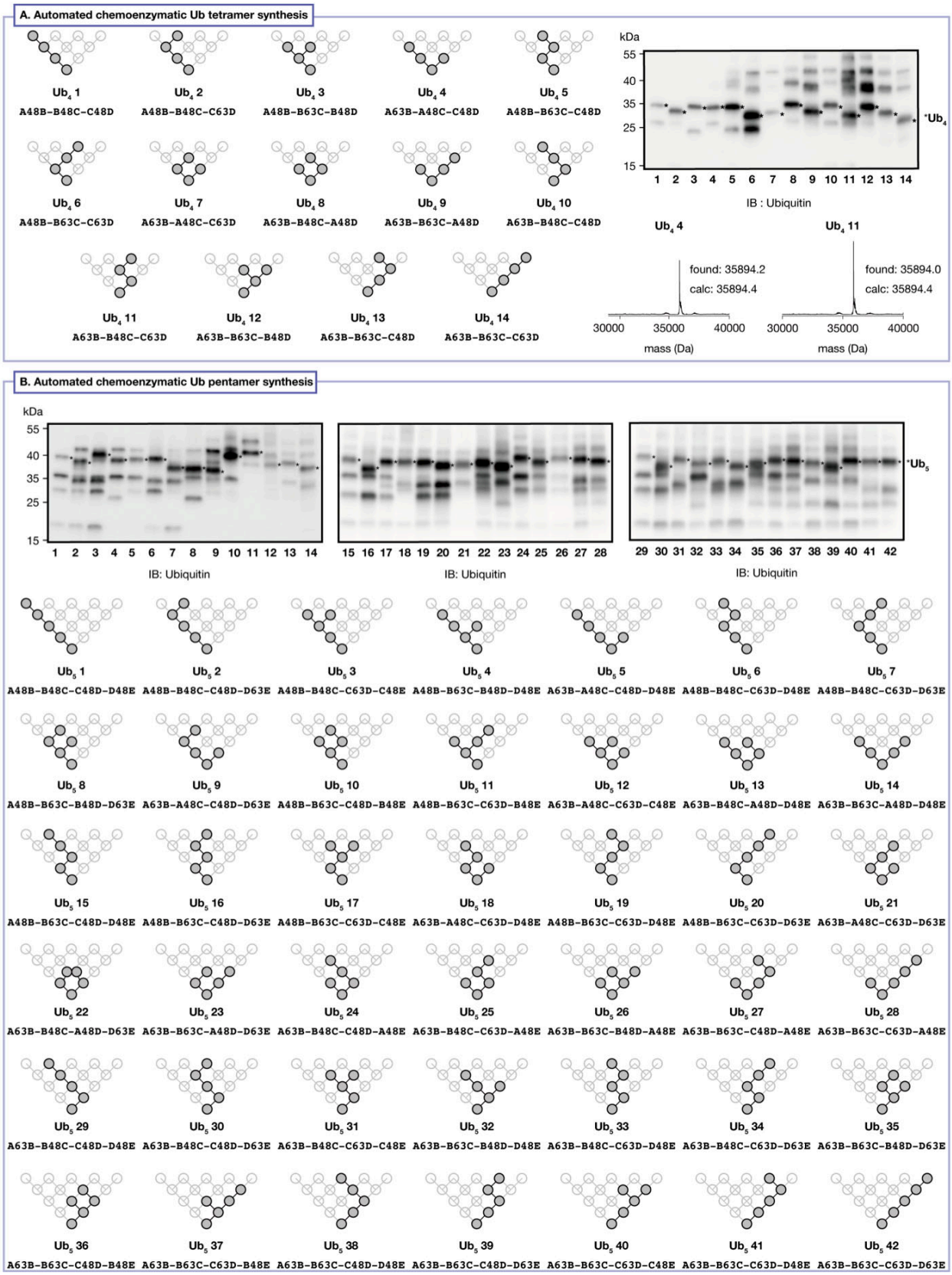


Figure 5. A) Western blot of 14 Ub tetramers (Aboc protected) prepared in a single run of the automated synthesis protocol. Each of the topologies are illustrated with the corresponding graph nomenclature. Two representative ESI-MS spectra confirming the tetramers are shown (see the Supplementary Information for the full dataset). B) Western blot of 42 K48/K63-linked Ub pentamers synthesized over three automated runs (14 chains per run). Each lane corresponds to a unique chain topology, shown below the blot using the graphical nomenclature. Bands at the expected molecular weight indicate successful synthesis of all topologies. All products retain Aboc protecting groups and a C-terminal His-tag.

https://doi.org/10.26434/chemrxiv-2025-943j7 **ORCID:** https://orcid.org/0000-0001-8394-8910 Content not peer-reviewed by ChemRxiv. **License:** CC BY-NC 4.0

Pentamers retaining Aboc masking groups and the C-terminal His-tag can be converted to the unprotected, unanchored Ub chains using our established procedure.[14] For example, after eluting from the magnetic Ni-NTA resin using imidazole-containing buffer, **Ub_5 3** and **Ub_5 10** (both His-tagged with 6 Aboc groups) were treated with 200 mM Na borate buffer pH 8.5 containing 2 mM $NaIO_4$ with 1 mM pyrrolidine at 37 °C for 30 min, followed by quenching with Tris and overnight incubation at 37 °C to effect global Aboc deprotection (Figure 6A). The His-tag was subsequently cleaved using the deubiquitinase YUH1,[13,44–46] and the resulting mixtures were purified by size-exclusion chromatography with 50 mM NH_4OAc to obtain pure Ub pentamers. The resulting unanchored **Ub_5 10** chain can by site-selectively conjugated to a specific lysine residue by Lysine Acylation using Conjugating Enzymes (LACE) to a target protein bearing a C-terminal LACE tag (IKQE) (Figure 6B).[47,48]

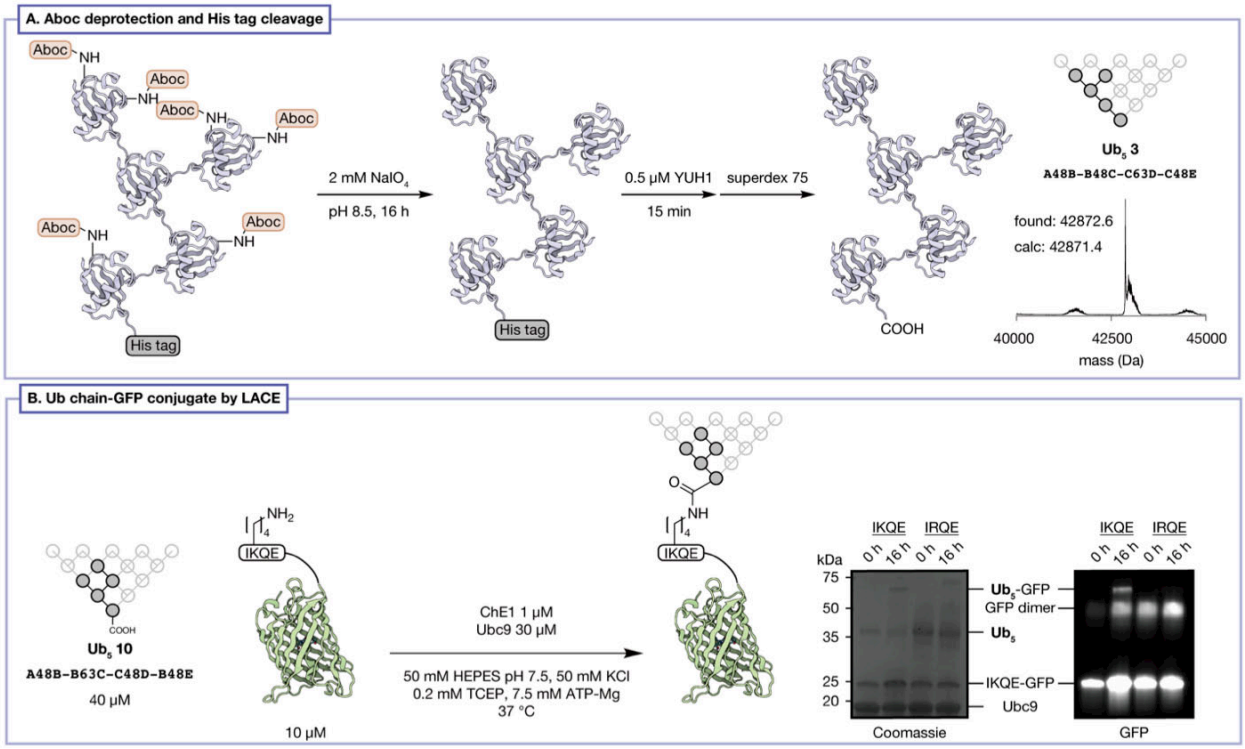


Figure 6. A) Deprotection scheme of Ub pentamers and obtained intact Ub pentamer mass of **Ub_5 3**. B) Site specific attachment of **Ub_5 10** to green fluorescent protein (GFP) using LACE.

Discussion

The production of homogenous, branched Ub chains has long posed a significant challenge. By combining innovative chemical advances – including a fast-cleaving protecting group for lysines on folded proteins and a scalable approach to Ub monomers – with a new digital representation of the complex Ub chain architectures, we established a robust, fully automatable approach to the comprehensive synthesis of all possible K48/K63 Ub tetramers and pentamers. Crucially, this

https://doi.org/10.26434/chemrxiv-2025-943j7 **ORCID:** https://orcid.org/0000-0001-8394-8910 Content not peer-reviewed by ChemRxiv. **License:** CC BY-NC 4.0

system does not merely automate a few synthetic operations but also digitally encodes the complete topological space of K48/K63 branched pentamers, simulates viable synthetic routes, and translates these into executable protocols for a widely accessible robotic platform.

While our focus here has been on K48/K63 linkages, the graph-based digital representation is extensible to other lysine linkages and topologies, which exceeds 10,000 possible structures for Ub pentamers alone. As new chemoenzymatic systems for accessing alternative branching sites emerge, and as protecting groups orthogonal to those employed here are identified, these innovations can be readily implemented into both the automation hardware and the underlying software platform. More broadly, this work illustrates how digitally encoded automation and site-specifically tailored synthetic biomolecules can converge to unlock previously inaccessible areas of biopolymer construction.

Safety Statement

No unexpected nor unusually high safety hazards were encountered.

Materials and Correspondance

The code used to reproduce the simulation and download associated application presented in this Article is available on GitHub (https://github.com/erickummelstedt/ubiquitinformatics.git). Extended data figures are available for the nomenclature description and the application of a graph-based representation of the Ub chains. Additional figures, methods, characterization data by NMR, IR, HPLC, mass spectrometry, and SDS-PAGE gels are found in the supplementary information. Plasmids generated for this study will be made available through Addgene (www.addgene.org/). Correspondence regarding any materials used in this study should be addressed to JWB.

Acknowledgment

This work was supported by the European Research Council Synergy Grant under the European Union's Horizon 2020 research and innovation program (No. 856581 – CHUbVi). S.M. acknowledges the Takenaka Scholarship Foundation for a scholarship. Celine Nardo (ETH Zürich) is thanked for peptide segment synthesis. The authors thank Patrick Matthias (FMI Basel, CH), and Yohei Yamauchi (ETH Zürich) for helpful discussions. We acknowledge the Molecular and Biomolecular Analysis Service (MoBiAS) at ETH Zürich for mass spectrometry and analysis.

https://doi.org/10.26434/chemrxiv-2025-943j7 **ORCID:** https://orcid.org/0000-0001-8394-8910 Content not peer-reviewed by ChemRxiv. **License:** CC BY-NC 4.0

Author Contributions

S.M. and E.K. contributed equally to this work. S.M., E.K., T.M., and J.W.B conceived the project. S.M., T.M., L.S., and J.H. conducted experimental work. S.M. wrote the automation scripts and performed the automated synthesis. E.K. wrote, designed and implemented the software. All authors contributed to data analysis. S.M., E.K., and J.W.B wrote the manuscript with contributions from the other authors.

References

1. Agrata, R. & Komander, D. Ubiquitin—A structural perspective. *Mol. Cell* **85**, 323–346 (2025).
2. Ohtake, F., Tsuchiya, H., Saeki, Y. & Tanaka, K. K63 ubiquitylation triggers proteasomal degradation by seeding branched ubiquitin chains. *Proceedings of the National Academy of Sciences* **115**, E1401–E1408 (2018).
3. Liu, C., Liu, W., Ye, Y. & Li, W. Ufd2p synthesizes branched ubiquitin chains to promote the degradation of substrates modified with atypical chains. *Nat. Commun.* **8**, 14274 (2017).
4. French, M. E., Koehler, C. F. & Hunter, T. Emerging functions of branched ubiquitin chains. *Cell Discovery* vol. 7 6 Preprint at https://doi.org/10.1038/s41421-020-00237-y (2021).
5. Banerjee, I. *et al.* Influenza A virus uses the aggresome processing machinery for host cell entry. *Science* **346**, 473–477 (2014).
6. Hao, R. *et al.* Proteasomes activate aggresome disassembly and clearance by producing unanchored ubiquitin chains. *Mol. Cell* **51**, 819–828 (2013).
7. Moreira, E. A., Yamauchi, Y. & Matthias, P. How influenza virus uses host cell pathways during uncoating. *Cells* **10**, 1722 (2021).
8. Yamauchi, Y. Chapter One - Influenza A virus uncoating. in *Advances in Virus Research* (eds. Kielian, M., Mettenleiter, T. C. & Roossinck, M. J.) vol. 106 1–38 (Academic Press, 2020).
9. Fang, R. *et al.* Episodic transport of protein aggregates achieves a positive size selectivity in aggresome formation. *Nat. Commun.* **16**, 7852 (2025).
10. Chen, Y.-H. *et al.* VPS34 K29/K48 branched ubiquitination governed by UBE3C and TRABID regulates autophagy, proteostasis and liver metabolism. *Nat. Commun.* **12**, 1322 (2021).
11. Hospenthal, M. K., Freund, S. M. V. & Komander, D. Assembly, analysis and architecture of atypical ubiquitin chains. *Nat. Struct. Mol. Biol.* **20**, 555–565 (2013).
12. McFarland, M. R. & Kulathu, Y. Emerging tools and methods to study cell signalling mediated by branched ubiquitin chains. *Biochem. Soc. Trans.* BST20253015 (2025).
13. Kiss, L., James, L. C. & Schulman, B. A. UbiREAD deciphers proteasomal degradation code of homotypic and branched K48 and K63 ubiquitin chains. *Mol. Cell* (2025) doi:10.1016/j.molcel.2025.02.021.

https://doi.org/10.26434/chemrxiv-2025-943j7 **ORCID:** https://orcid.org/0000-0001-8394-8910 Content not peer-reviewed by ChemRxiv. **License:** CC BY-NC 4.0

14. Mikami, T., Majima, S., Song, H. & Bode, J. W. Biocompatible Lysine Protecting Groups for the Chemoenzymatic Synthesis of K48/K63 Heterotypic and Branched Ubiquitin Chains. *ACS Cent. Sci.* **9**, 1633–1641 (2023).
15. Winkler, D. F. H. Automated solid-phase peptide synthesis. *Methods Mol. Biol.* **2103**, 59–94 (2020).
16. Sandahl, A. F. *et al.* On-demand synthesis of phosphoramidites. *Nat. Commun.* **12**, 2760 (2021).
17. Crawford, C. J. *et al.* Automated Synthesis of Algal Fucoidan Oligosaccharides. *J. Am. Chem. Soc.* (2024) doi:10.1021/jacs.4c02348.
18. Guberman, M. & Seeberger, P. H. Automated Glycan Assembly: A perspective. *J. Am. Chem. Soc.* **141**, 5581–5592 (2019).
19. Yao, W. *et al.* Automated solution-phase multiplicative synthesis of complex glycans up to a 1,080-mer. *Nature Synthesis* 1–10 (2022).
20. Li, J. *et al.* Synthesis of many different types of organic small molecules using one automated process. *Science* **347**, 1221–1226 (2015).
21. Kowalski, D. J., MacGregor, C. M., Long, D.-L., Bell, N. L. & Cronin, L. Automated library generation and serendipity quantification enables diverse discovery in coordination chemistry. *J. Am. Chem. Soc.* **145**, 2332–2341 (2023).
22. Liu, J. & Hein, J. E. Automation, analytics and artificial intelligence for chemical synthesis. *Nat. Synth.* **2**, 464–466 (2023).
23. Li, T. *et al.* An automated platform for the enzyme-mediated assembly of complex oligosaccharides. *Nat. Chem.* **11**, 229–236 (2019).
24. Toma-Fukai, S. & Shimizu, T. Structural Diversity of Ubiquitin E3 Ligase. *Molecules* **26**, (2021).
25. Wang, Y. S., Wu, K. P., Jiang, H. K., Kurkute, P. & Chen, R. H. Branched Ubiquitination: Detection Methods, Biological Functions and Chemical Synthesis. *Molecules* **25**, 1–20 (2020).
26. Lange, S. M. *et al.* VCP/p97-associated proteins are binders and debranching enzymes of K48–K63-branched ubiquitin chains. *Nat. Struct. Mol. Biol.* 1–16 (2024).
27. Kumar, K. S. A. *et al.* Total chemical synthesis of a 304 amino acid K48-linked tetraubiquitin protein. *Angew. Chem. Weinheim Bergstr. Ger.* **123**, 6261–6265 (2011).
28. Tang, S. *et al.* Practical Chemical Synthesis of Atypical Ubiquitin Chains by Using an Isopeptide-Linked Ub Isomer. *Angewandte Chemie - International Edition* **56**, 13333–13337 (2017).
29. Furuhata, T., Devadasan Racheal, P. A., Murayama, I., Toyoda, U. & Okamoto, A. One-Pot, Photocontrolled Enzymatic Assembly of the Structure-Defined Heterotypic Polyubiquitin Chain. *J. Am. Chem. Soc.* **145**, 11690–11700 (2023).
30. Harmand, T. J., Murar, C. E. & Bode, J. W. Protein chemical synthesis by α-ketoacid-hydroxylamine ligation. *Nat. Protoc.* **11**, 1130–1147 (2016).

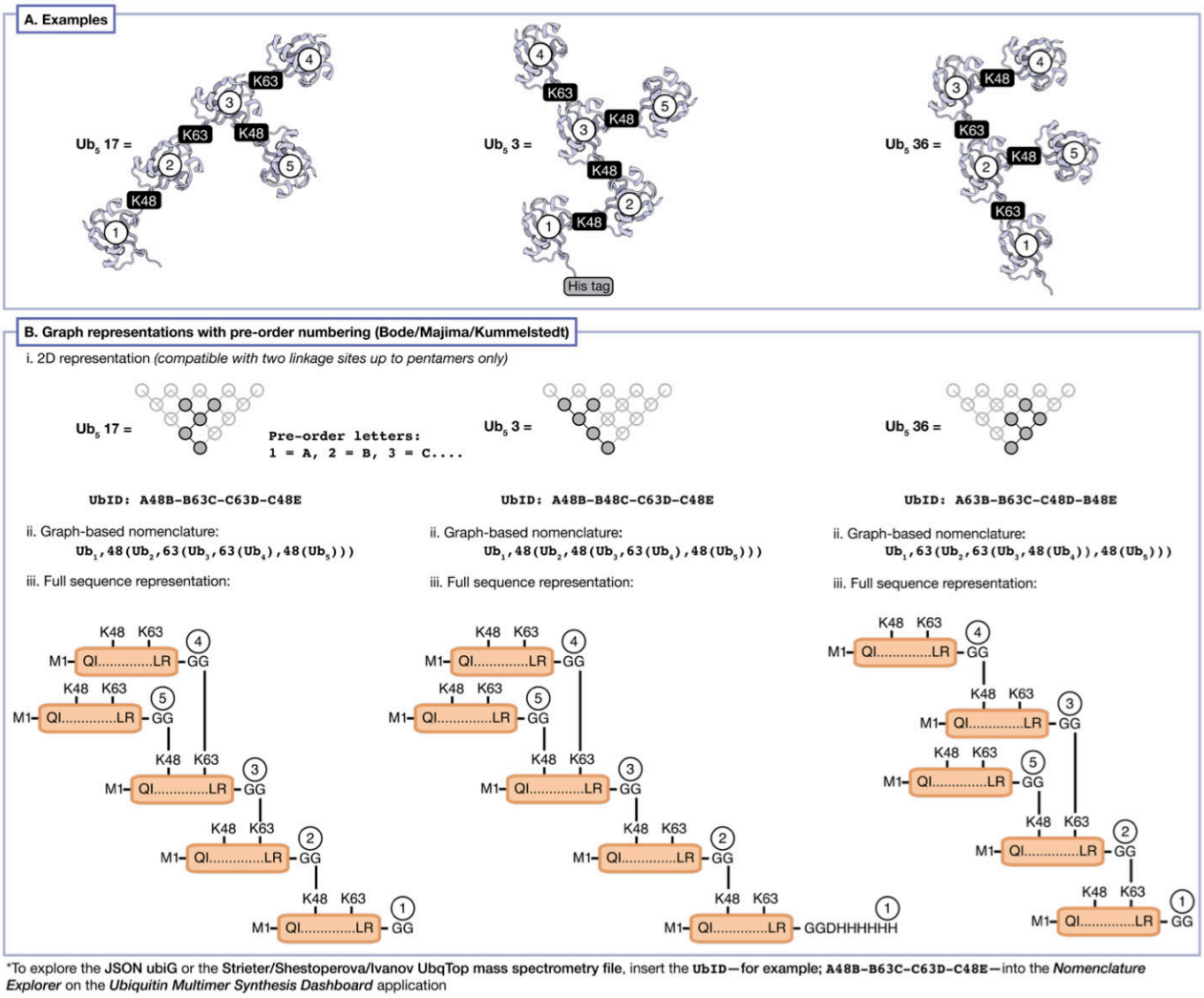
https://doi.org/10.26434/chemrxiv-2025-943j7 **ORCID:** https://orcid.org/0000-0001-8394-8910 Content not peer-reviewed by ChemRxiv. **License:** CC BY-NC 4.0

31. Han, J. *et al.* Cyclic Hydroxylamines for Native Residue–Forming Peptide Ligations: Synthesis of Ubiquitin and Tirzepatide. 10.1021/jacs.5c11881. *Manuscript in press.*
32. El Oualid, F. *et al.* Chemical Synthesis of Ubiquitin, Ubiquitin-Based Probes, and Diubiquitin. *Angew. Chem. Int. Ed.* **49**, 10149–10153 (2010).
33. van Tilburg, G. B. A. *et al.* K27-Linked Diubiquitin Inhibits UCHL3 via an Unusual Kinetic Trap. *Cell Chemical Biology* **28**, 191-201.e8 (2021).
34. Dong, K. C. *et al.* Preparation of distinct ubiquitin chain reagents of high purity and yield. *Structure* **19**, 1053–1063 (2011).
35. Blythe, E. E., Olson, K. C., Chau, V. & Deshaies, R. J. Ubiquitin- and ATP-dependent unfoldase activity of P97/VCP•NPLOC4•UFD1L is enhanced by a mutation that causes multisystem proteinopathy. *Proc. Natl. Acad. Sci. U. S. A.* **114**, E4380–E4388 (2017).
36. El Hebieshy, A. F. *et al.* Site-directed multivalent conjugation of antibodies to ubiquitinated payloads. *Nat. Biomed. Eng.* 1–16 (2025).
37. Pluska, L. *et al.* The UBA domain of conjugating enzyme Ubc1/Ube2K facilitates assembly of K48/K63-branched ubiquitin chains. *EMBO J.* **40**, e106094 (2021).
38. During the preparation of this manuscript, another graph-based representation of Ub chains was posted: Shestoperova, E. I., Ivanov, D., Zhong, J., Chien, L. & Strieter, E. Computationally driven top-down mass spectrometry of ubiquitinated proteins. *bioRxiv* (2025) doi:10.1101/2025.07.24.666707.
39. *Fundamental Algorithms 2nd Edn*. vol. 1 (Addison-Wesley, 1973).
40. Mehta, A. Y. & Cummings, R. D. GlycoGlyph: a glycan visualizing, drawing and naming application. *Bioinformatics* **36**, 3613–3614 (2020).
41. Geis-Asteggiante, L., Lee, A. E. & Fenselau, C. Analysis of the topology of ubiquitin chains. *Methods Enzymol.* **626**, 323–346 (2019).
42. Lee, A. E. *et al.* Preparing to read the ubiquitin code: top-down analysis of unanchored ubiquitin tetramers. *J. Mass Spectrom.* **51**, 629–637 (2016).
43. Moore, E. F. shortest path through maze. in *Proc. Int. Symp. Theory Switching* 285–292 (Harvard Univ. Press, 1959).
44. Pickart, C. M. & Rose, I. A. Ubiquitin carboxyl-terminal hydrolase acts on ubiquitin carboxyl-terminal amides. *J. Biol. Chem.* **260**, 7903–7910 (1985).
45. Raasi, S. & Pickart, C. M. Ubiquitin chain synthesis. *Methods Mol. Biol.* **301**, 47–55 (2005).
46. Castañeda, C. A. *et al.* Controlled enzymatic synthesis of natural-linkage, defined-length polyubiquitin chains using lysines with removable protecting groups. *Chem. Commun.* **47**, 2026–2028 (2011).
47. Hofmann, R., Akimoto, G., Wucherpfennig, T. G., Zeymer, C. & Bode, J. W. Lysine acylation using conjugating enzymes for site-specific modification and ubiquitination of recombinant proteins. *Nat. Chem.* **12**, 1008–1015 (2020).

https://doi.org/10.26434/chemrxiv-2025-943j7 **ORCID:** https://orcid.org/0000-0001-8394-8910 Content not peer-reviewed by ChemRxiv. **License:** CC BY-NC 4.0

48. Akimoto, G., Fernandes, A. P. & Bode, J. W. Site-Specific Protein Ubiquitylation Using an Engineered, Chimeric E1 Activating Enzyme and E2 SUMO Conjugating Enzyme Ubc9. *ACS Central Science* acscentsci.1c01490 (2022).

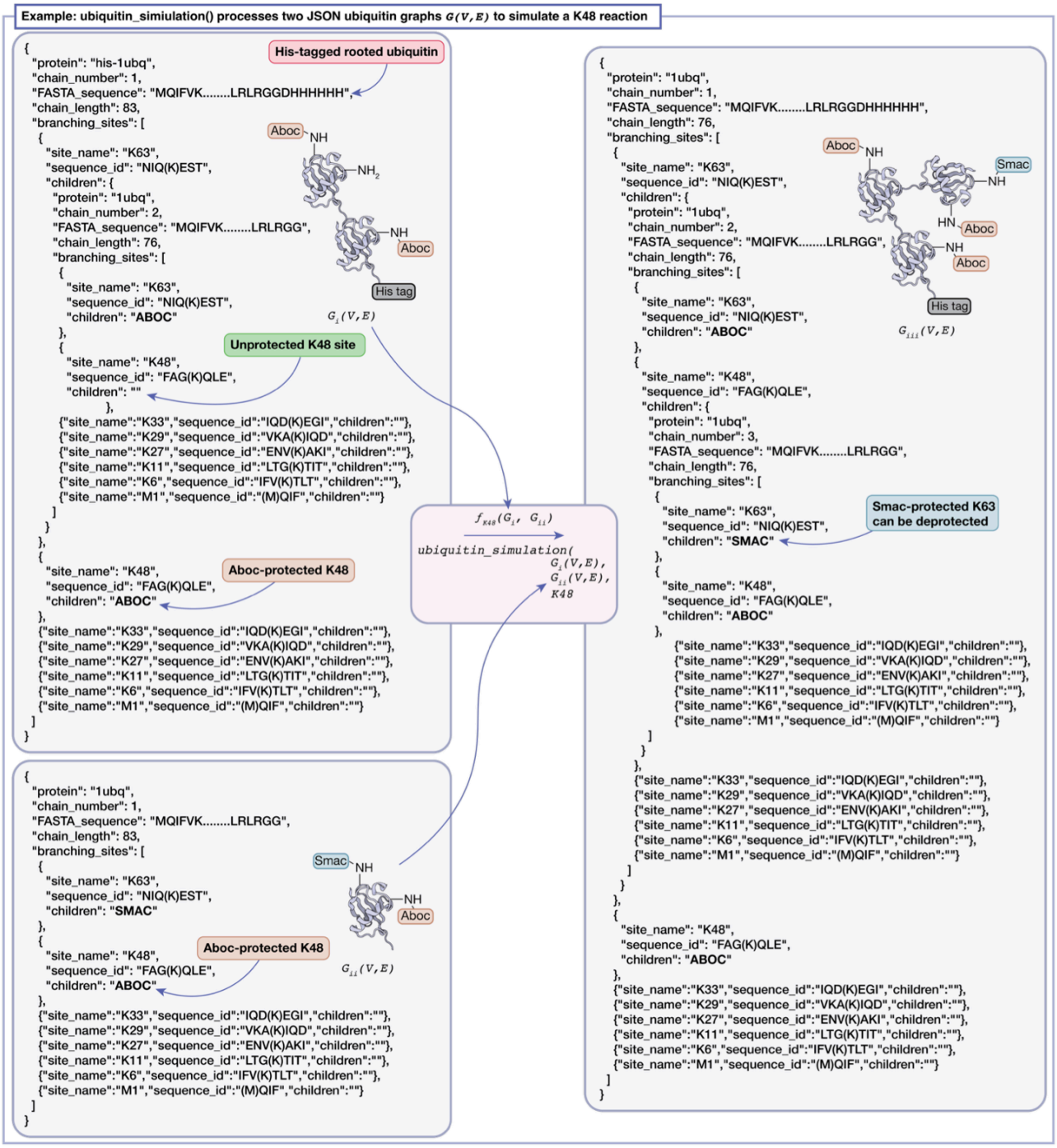
https://doi.org/10.26434/chemrxiv-2025-943j7 **ORCID:** https://orcid.org/0000-0001-8394-8910 Content not peer-reviewed by ChemRxiv. **License:** CC BY-NC 4.0



*To explore the JSON ubiG or the Strieter/Shestoperova/Ivanov UbqTop mass spectrometry file, insert the `UbID`—for example; `A48B-B63C-C63D-C48E`—into the *Nomenclature Explorer* on the *Ubiquitin Multimer Synthesis Dashboard* application

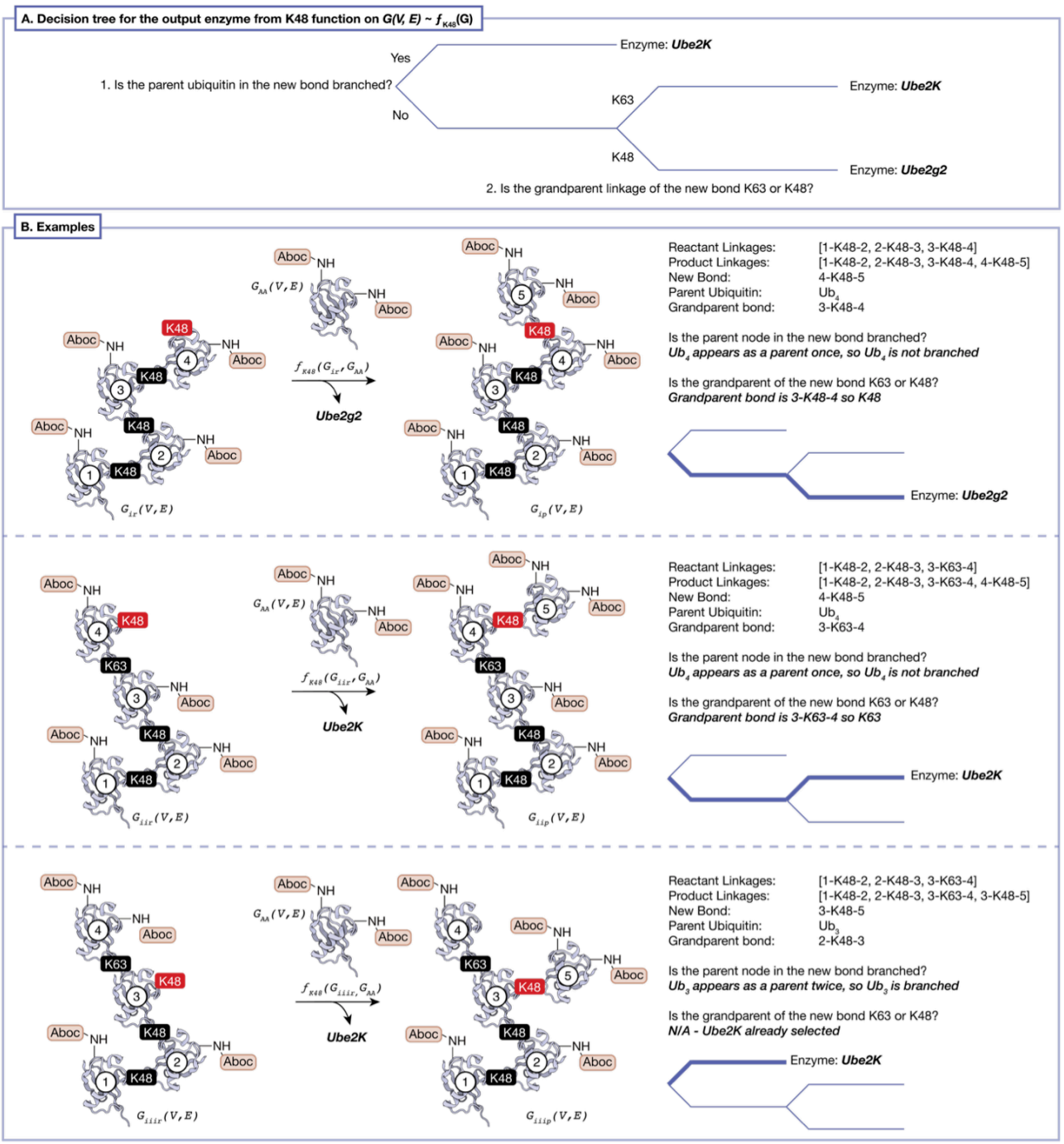
Extended Data Figure 1. Nomenclature of three ubiquitin chains: Ub_5 17, Ub_5 3, and Ub_5 36. Shown are: (A) cartoon depictions of the chain structures, (Bi) 2D representations are compatible with species up to pentamers containing two linkage sites; (Bii) a graph-based system, termed UbID, derived from pre-order numbering where letters represent node indices. UbID strings can be entered into the Nomenclature Explorer tab of the downloadable application (GitHub: https://github.com/erickummelstedt/ubiquitinformatics.git); and (Biii) a graph-based nomenclature combining pre-order numbering with a full sequence representation to showcase biological directionality of the pre-order logic.

https://doi.org/10.26434/chemrxiv-2025-943j7 ORCID: https://orcid.org/0000-0001-8394-8910 Content not peer-reviewed by ChemRxiv. License: CC BY-NC 4.0



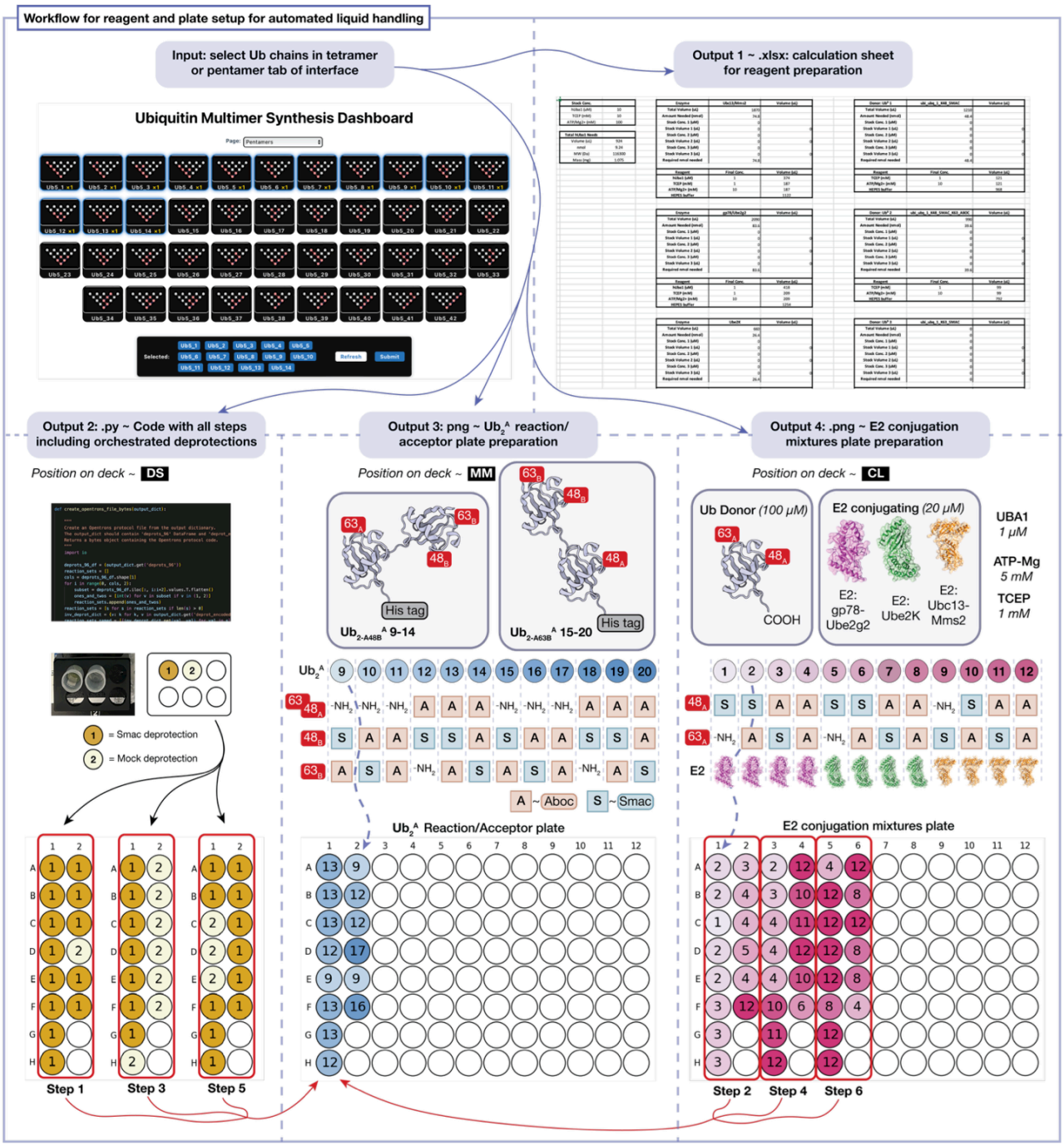
Extended Data Figure 2. Full JSON representation of a ubiquitin conjugation via K48 linkage. The example illustrates how the His-tag, Aboc, and Smac protecting groups are encoded within the JSON, and how a ubiquitin–ubiquitin conjugation is represented as a nested structure. Each ubiquitin node stores protein identification, its chain number (assigned by pre-order traversal), the FASTA sequence of the vertex, the sequence length, and the defined branching sites. Each branching site is annotated with the site name, its amino acid motif, and any children nodes corresponding to conjugated species at that edge. The simulation engine operates through functions applied to these JSON-encoded graphs. Shown is the example of the ubiquitin_simulation function evaluating how two ubiquitins conjugate under specific conditions (here, a K48 reaction).

https://doi.org/10.26434/chemrxiv-2025-943j7 **ORCID:** https://orcid.org/0000-0001-8394-8910 Content not peer-reviewed by ChemRxiv. **License:** CC BY-NC 4.0



Extended Data Figure 3. Decision logic for K48 enzyme assignment in function: assign_correct_E2_enzyme. The function compares the edges of the product ubiG with those of the reactant, identifying the newly formed bond, and evaluates its local environment using the decision tree shown. If the parent node of the new bond becomes branched, the enzyme is assigned as Ube2K. If not, the function inspects the grandparent edge: a K63 linkage directs assignment to Ube2K, whereas a K48 linkage assigns Ube2g2.

https://doi.org/10.26434/chemrxiv-2025-943j7 **ORCID:** https://orcid.org/0000-0001-8394-8910 Content not peer-reviewed by ChemRxiv. **License:** CC BY-NC 4.0



Extended Data Figure 4. Reagent and plate preparation for automated synthesis of Ub_5 1 to Ub_5 14. The input is a customized selection of K48/K63 tetramers or pentamers from the application (GitHub: https://github.com/erickummelstedt/ubiquitinformatics.git), which generates four outputs: an .xlsx calculation sheet detailing reagent preparation, including premixing of enzyme and donor solutions into stock E2 conjugation mixtures (E1 1 µM, E2 20 µM, Ub^D 100 µM, 2 equiv, Mg-ATP 5 mM, TCEP 1 mM) and the required amounts of Ub_2^A dimers, 10 nmol per well (with some dimers the starting species across pathways); a .py script to run the liquid-handling robot, including on-deck deprotections from designated wells (with a .png illustration of the deprotection additions); a reaction plate map (.png) showing the **Ub_2^A** dimer positions on the 96-well plate for the magnetic module; and a conjugation mixture plate map (.png) specifying the positioning of the mixtures in glass vials in the 4 °C cooling rack. The sequence of additions from the E2 conjugation mixture plate and deprotection solutions is shown at the bottom.

https://doi.org/10.26434/chemrxiv-2025-943j7 **ORCID:** https://orcid.org/0000-0001-8394-8910 Content not peer-reviewed by ChemRxiv. **License:** CC BY-NC 4.0